

CS 6501 — Week 4 Cheat Sheet: Safety, Liveness, and Deadlock

Big Picture

Week 4 focuses on the distinction between **safety** and **liveness** in multi-robot systems. While safety ensures that “nothing bad happens,” liveness ensures that “something good eventually happens.” The week highlights how many safe control methods fail in practice due to deadlock, freezing, or livelock, motivating new approaches to balancing safety and progress.

Safety in Multi-Robot Systems

Safety typically means maintaining the system state within a predefined **safe set**:

$$\mathcal{C} = \{x \mid h(x) \geq 0\}$$

If the system starts in the safe set and respects appropriate constraints, it should remain safe for all future time.

Control Barrier Functions (CBFs)

A Control Barrier Function $h(x)$ enforces safety by restricting control inputs so that:

$$\dot{h}(x, u) \geq -\alpha(h(x))$$

where $\alpha(\cdot)$ is a class- $\mathcal{K}$ function.

Key properties:

- CBFs enforce forward invariance of the safe set
- Safety is guaranteed by construction
- Control inputs are minimally modified to satisfy constraints

CBFs for Multi-Robot Systems

In multi-robot systems, safety constraints are imposed pairwise:

$$h_{ij}(x) = \|p_i - p_j\|^2 - d_{\min}^2$$

This leads to:

- A large number of constraints
- Coupled control decisions
- Increased conservatism as robot density grows

High-Order and Neural CBFs

High-Order CBFs (HOCBFs) extend barrier methods to higher-order dynamics.

Neural CBFs / BarrierNet:

- Learn barrier functions from data
- Embed safety constraints inside neural network architectures
- Provide end-to-end differentiability

BarrierNet Theorem (informal): If the system state starts in the intersection of all learned safe sets, then it remains in that safe set for all time under the learned controller.

Reliability vs Conservatism

CBF-based systems are:

- Highly reliable for safety
- Often overly conservative

As more constraints activate, feasible control actions shrink, leading to stalled behavior.

Liveness

Liveness ensures that agents eventually make progress toward their goals.

Formally:

- Safety: nothing bad ever happens
- Liveness: something good eventually happens

CBFs guarantee safety, but **do not guarantee liveness**.

Freezing Robot Problem (FRP)

The Freezing Robot Problem occurs when:

- All robots remain safe
- No robot can move without violating safety
- The system reaches a deadlock or livelock

This often occurs near:

- Goal regions
- Narrow passages
- Highly symmetric configurations

Deadlock and Livelock

Deadlock:

- Robots stop permanently
- No progress is possible

Livelock:

- Robots keep moving
- No robot reaches its goal

Both satisfy safety but violate liveness.

Symmetry as a Root Cause

Deadlock often arises from symmetry:

- Identical robots
- Identical goals or priorities
- Symmetric environments

As relative velocity vectors flatten, robots mirror each other's actions and stall.

Deadlock Resolution Strategies

Approaches to restore liveness include:

- Priority assignment
- Mode switching
- Heuristic rules (e.g., right-hand rule)
- Breaking symmetry explicitly

These methods trade optimality or fairness for progress.

Zhou et al. (2017): Buffered Voronoi Cells

Zhou et al. propose:

- Fast, online collision avoidance
- Buffered Voronoi partitions of space
- Local reasoning without centralized coordination

Each robot selects motion based on nearest neighbors, with buffers ensuring safety.

Key Tradeoff: Safety vs Liveness

- Strong safety guarantees increase conservatism
- Conservatism reduces liveness
- Practical systems must balance both

This motivates hybrid methods combining planning, control, and coordination.

Relation to the Course

Week 4 highlights:

- The limits of purely reactive safety
- Why guarantees alone are insufficient
- The need for structure, priorities, or planning layers

This sets up future topics in:

- Liveness guarantees
- Social compliance
- Game-theoretic and learning-based coordination

One-Sentence Summary

Week 4 shows that while Control Barrier Functions guarantee safety, they often fail to ensure liveness, leading to deadlock and freezing behavior in symmetric multi-robot systems.